EXPERIMENT NO. 20:

Implement Future Sales Prediction using a suitable machine learning algorithm

Aim:

To implement a **Future Sales Prediction Model** in Python using **Linear Regression** and predict future sales based on previous sales data.

Procedure:

1. Import the required Python libraries.
2. Create a sample sales dataset.
3. Separate the input feature and target sales values.
4. Create and train the Linear Regression model.
5. Predict sales for future periods.
6. Display the predicted future sales values.

Program:

```
# Future Sales Prediction using Linear Regression

import numpy as np

from sklearn.linear_model import LinearRegression

# Previous sales data

months = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]).reshape(-1, 1)

sales = np.array([120, 135, 150, 160, 175, 190, 205, 220, 235, 250, 265, 280])

# Create Linear Regression model

model = LinearRegression()

# Train the model

model.fit(months, sales)

# Predict future sales

future_months = np.array([13, 14, 15, 16, 17, 18]).reshape(-1, 1)
```

```
future_sales = model.predict(future_months)

print("Future Months:")

print(future_months.flatten())

print("\nPredicted Future Sales:")

print(future_sales.round(2))
```

Output:

```
Future Months:
[13 14 15 16 17 18]

Predicted Future Sales:
[293.18 307.71 322.24 336.77 351.29 365.82]
```

Result:

The **Future Sales Prediction Model** was successfully implemented using **Linear Regression**. The model learned the trend from previous sales data and successfully predicted the expected sales for future months.